Area-weighted historical population

Control: Match of Historical to Modern Geographic Units — Area-weighted population. At State level, this is a historical-instrument choice.

1. Allocate population using geographic overlap. Let P_{jt} be the population of historical reporting area j in census year t, and a_{jt} its mapped area. Most reporting areas are counties. Let $B_{jg,t}$ be the part inside target region g. The allocated historical population is

$$P_{g,t}^{\text{area}} = \sum_j P_{jt} \frac{B_{jg,t}}{a_{jt}}.$$

An MSA is the union of its current member counties. In State analysis, a region is the union of modern counties assigned to one historical FE group in the dataset. The territory year fixes this grouping; the instrument year selects P_{jt} and its source areas. County matching uses the same target membership. The methods differ in population allocation.

2. The assumption is uniform density within each reporting area. If 30% of a historical county's area lies inside an MSA, the method assigns 30% of that county's population to the MSA. It does not know where people lived within that county. For example, 30% of an area with 10,000 residents contributes 3,000 residents. The method therefore handles boundary crossings mechanically, but can misallocate population if settlement was concentrated on one side of the boundary. It uses geographic area rather than modern population weights.

3. The result is a population total. Under Levels, $z_{g,t} = P_{g,t}^{\text{area}}/1{,}000$ is in thousands of people. Natural log uses $\log P_{g,t}^{\text{area}}$ for positive totals. The total instruments modern employment or employment density, according to the selected approach. It is not divided by modern area. It also differs from Weighted Density Overlap, which averages historical county densities. The minimum-overlap threshold used by the overlap instruments does not apply here.

4. Reporting and geographic coverage matter. Only historical sources in the selected states or states-and-territories scope contribute. Joint county reports use their combined area and count the reported total once. No mapped overlap, unknown population in an overlapping source, or a flagged source omission can make the total unavailable. A recorded zero remains zero. Some historical records lack usable polygons; where their locations are known, affected totals are flagged. Otherwise, County and MSA totals cover mapped records only. State totals additionally require at least 95% geographic coverage. This threshold is not a guarantee that the uniform-density assumption is accurate.